

Unidad X: Gravitación

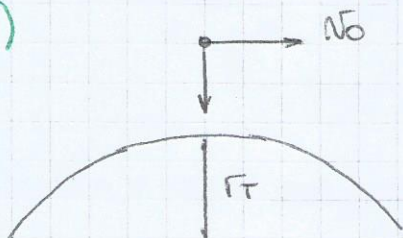
- Galileo - Galilei (1564 - 1642)
- Johannes Kepler (1571 - 1630)
- Sir Isaac Newton (1642 - 1727)

a) $F = m \cdot a$

$$a = g$$

$$P \propto m$$

b)



$$a_L = v_L^2 / R_L$$

a_L = aceleración centrípeta

R_L = radio de órbita lunar

$$a_p = g = \frac{4\pi^2 r_T}{T_p^2}$$

$$\frac{a_L}{a_p} = \frac{R_L}{R_T} \cdot \frac{T^2}{T_L^2} \neq 1$$

T_L = periodo lunar

$$v_L = r_L \omega_L = r_L \frac{2\pi}{T_L}$$

$$a_L = \frac{v_L^2}{R_L} = \frac{4\pi^2 R_L}{T_L^2}$$

c) $T^2 \propto R^3$ (3ª ley de Kepler)

$$\frac{T^2}{T_L^2} = \frac{R_T^3}{R_L^3}$$

$$\frac{g_L}{g} = \frac{R_L}{R_T} \cdot \frac{R_T^3}{R_L^3} = \frac{R_T^2}{R_L^2}$$

$$g_L R_L^2 = g R_T^2 = \text{cte}$$

$$g(r) = \frac{\text{cte}}{r^2}$$

$$F_{T,L} = m_L g_L = \frac{K m_L}{R_L^2}$$

3^{ra} Ley de Newton: $F_{T,L} = -F_{L,T}$

$$F_{L,T} = m_T g_T = \frac{K m_T}{R_L^2}$$

$F_{L,T}$ = Proporcional a m_T

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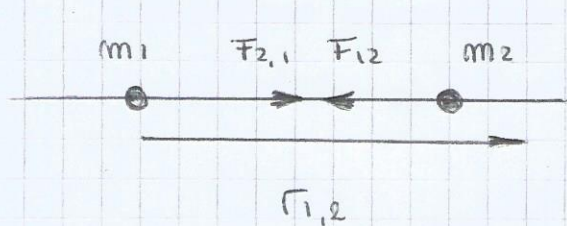
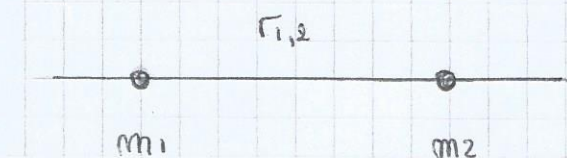
$F_{L,T}$ = Inv. proporcional a R_L^2

$$\frac{K m_L}{R_L^2} = - \frac{K m_T}{R_L^2}$$

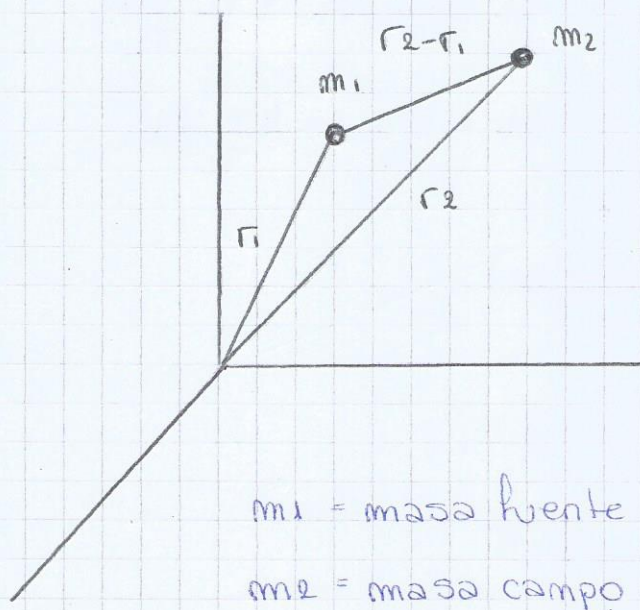
Ley de gravitación universal:

$$F_{1,2} = - \frac{G \cdot m_1 \cdot m_2}{(r_2 - r_1)^2} \hat{r}_{1,2}$$

$$G = 6,672 \times 10^{-11} \frac{m^3}{kg \cdot s^2}$$



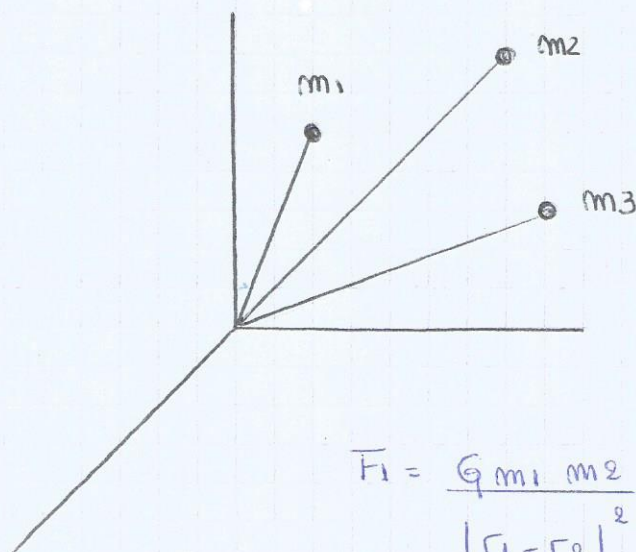
$$F_{1,2} = - \frac{G m_1 m_2}{r_{1,2}^2} \hat{r}_{1,2} = -F_{2,1}$$



$$F_{1,2} = - \frac{G m_1 m_2}{|r_2 - r_1|^3} (r_2 - r_1)$$

$$\hat{r}_{1,2} = \frac{(r_2 - r_1)}{|r_2 - r_1|}$$

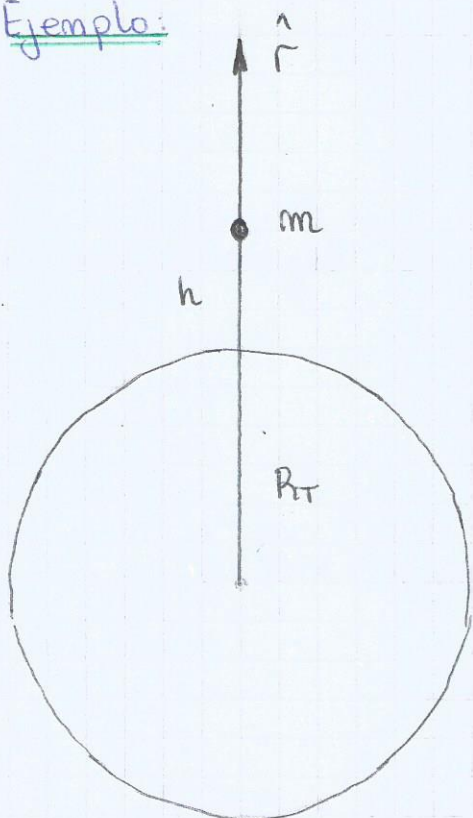
Principio de superposición de fuerzas:



$$F_1 = \frac{G m_1 m_2}{|r_1 - r_2|^2} \hat{r}_{1,2} + \frac{G m_1 m_3}{|r_1 - r_3|^2} \hat{r}_{1,3}$$

$$F_i = \sum_{j=1}^N \frac{G m_i m_j}{|r_i - r_j|^3} (r_i - r_j)$$

Ejemplo:



Si $R_T \gg h$

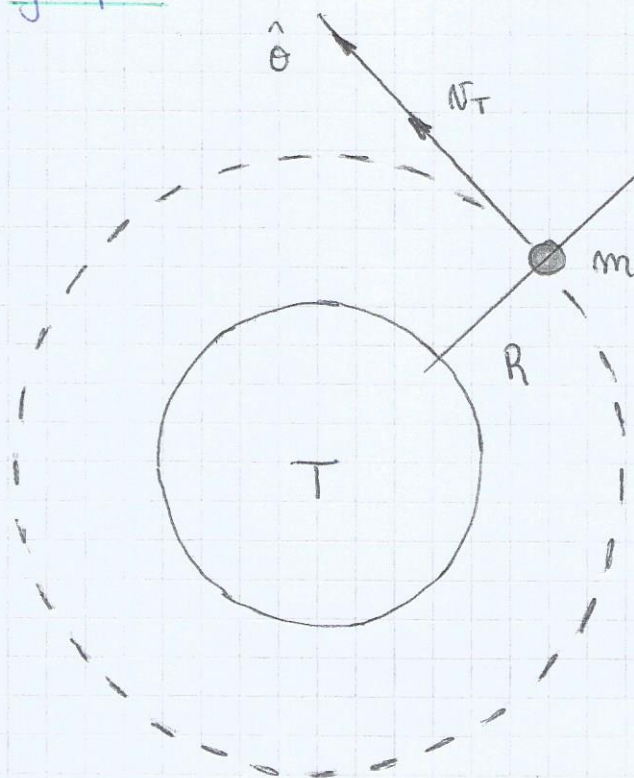
$$F = \frac{-G m M_T}{(R_T + h)^2} \hat{r} = -m g(r) \hat{r}$$

$$g(r) = \frac{G M_T}{(R_T + h)^2} = \frac{G M_T}{R_T^2 \left(1 + \frac{h}{R_T}\right)^2}$$

$$\approx \frac{G M_T}{R_T^2} \left(1 - \frac{2h}{R_T} + \frac{3h^2}{R_T^2} - \dots\right)$$

$$g(r) = g_0 \left(1 - \frac{2h}{R_T} + \frac{3h^2}{R_T^2} - \dots\right) \text{ Con } g_0 = \frac{G M_T}{R_T^2}$$

Ejemplo:



$$\hat{r}) - \frac{G M m}{R^2} \hat{r} = - \frac{m v_0^2}{R} \hat{r}$$

$$\hat{\theta}) 0 = m R \ddot{\theta} \Rightarrow \dot{\theta} = 0 \quad \omega = \text{cte}$$

$$R(v_0) = \frac{G M}{v_0^2}$$

Para cada v_0 hay un único radio para el cual el proyectil puede realizar órbita circular.

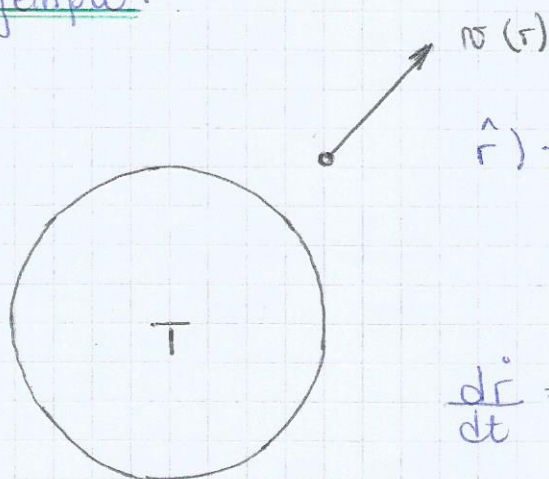
$$\left\{ \begin{array}{l} \omega = \frac{2\pi}{T} \\ v_0 = \omega R \end{array} \right.$$

$$\frac{G M}{R^2} = \omega^2 R$$

$$\frac{G M}{R^2} = \frac{4\pi^2}{T^2} R$$

$$T^2 = \frac{4\pi^2}{G M} R^3 \quad 3^{\text{ra}} \text{ ley de Kepler}$$

Ejemplo:



$$\hat{r}) - \frac{G M m}{r^2} \hat{r} = m (\ddot{r} - r \ddot{\theta}) \hat{r}$$

$$\ddot{r} = - \frac{G M}{r^2}$$

$$\frac{dr}{dt} = \frac{dr}{dr} \frac{dr}{dt} = \frac{dr}{dr} \dot{r} = - \frac{G M}{r^2}$$

$$\int_{v_0}^{v(r)} \dot{r} dr = -GM \int_{r_0}^r \frac{1}{r^2} dr$$

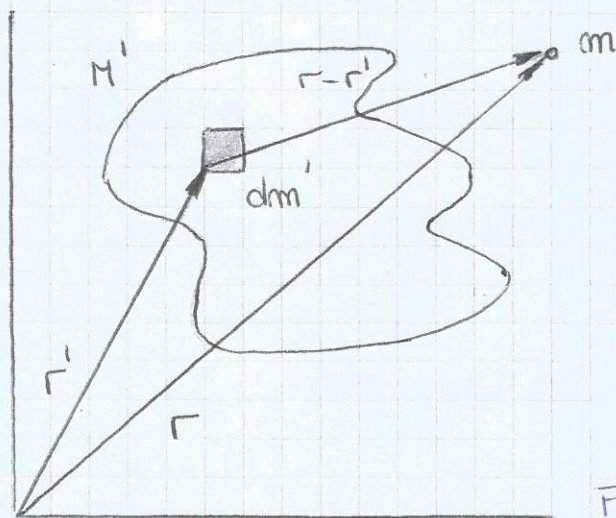
$$v(r) = \sqrt{v_0^2 - 2GM \left(\frac{1}{r_0} - \frac{1}{r} \right)}$$

$$v_0 - 2GM \left(\frac{1}{r_0} - \frac{1}{r_{\max}} \right) = 0$$

$$r_{\max} = \frac{R_T}{1 - \frac{v_0^2}{2GM} \cdot R_T} \quad (\text{Radio máximo}).$$

Velocidad de escape

$$v_e = \lim_{r_{\max} \rightarrow \infty} \left[2GM \left(\frac{1}{R_T} - \frac{1}{r_{\max}} \right) \right]^{\frac{1}{2}} = \sqrt{\frac{2GM}{R_T}} \approx 11,2 \frac{\text{km}}{\text{s}}$$



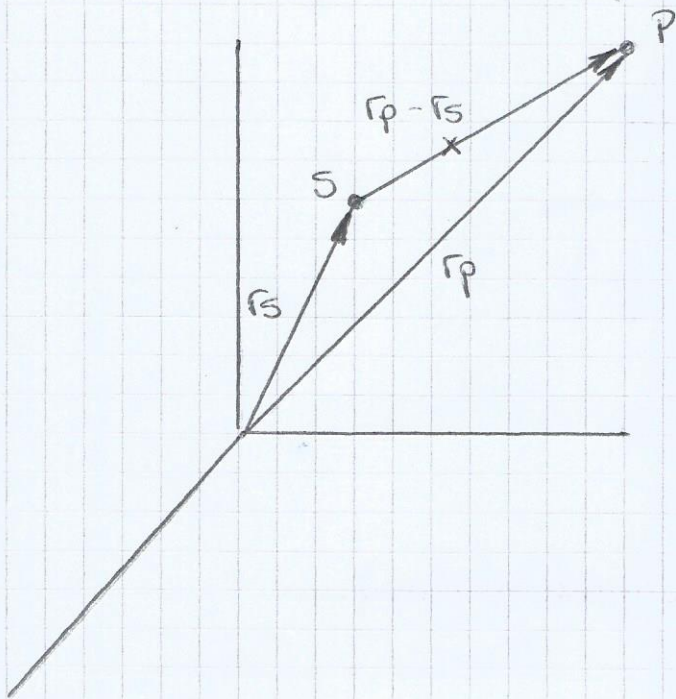
$$dF = \frac{-G m dm' (r-r')}{|r-r'|^3}$$

$$F = \int_M \frac{-G m dm' (r-r')}{|r-r'|^3}$$

$$F = \int_{V'} \frac{-GM}{|r-r'|^3} (r-r') g'(r') dv'$$

$$\text{Con } g'(r') = \frac{dm'}{dv'}$$

El problema de Kepler:

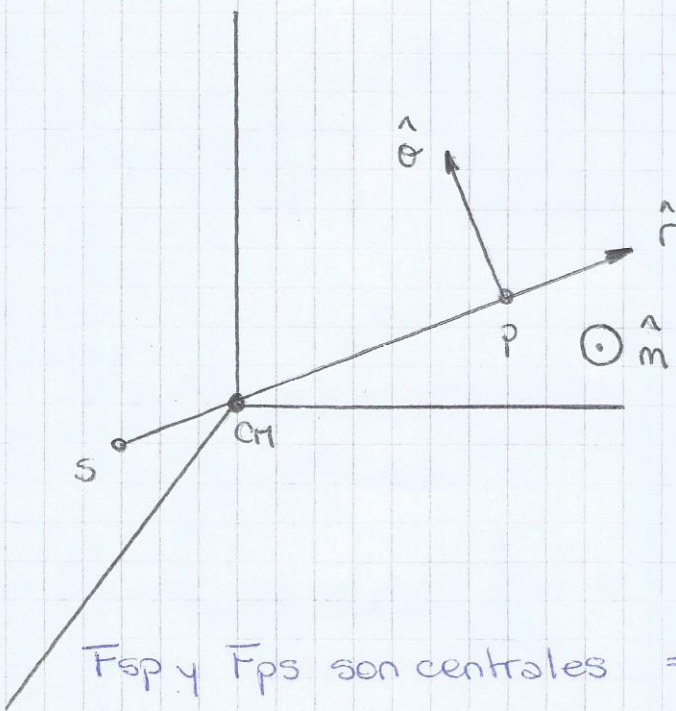


$$\mathbf{r} = \mathbf{r}_p - \mathbf{r}_s$$

$$\mathbf{R}_{cm} = \frac{M \mathbf{r}_s + m \mathbf{r}_p}{M + m}$$

$$\ddot{\mathbf{R}}_{cm} = 0 \Rightarrow \text{MRU}$$

$$\mathbf{R}_{cm} = \mathbf{R}_{cm_0} + \mathbf{V}_{cm} t$$



$$\mathbf{r}_p = \mathbf{R}_{cm} + \frac{M}{m+M} \mathbf{r}$$

$$\mathbf{r}_s = \mathbf{R}_{cm} - \frac{m}{m+M} \mathbf{r}$$

$$\mathbf{F} = \mu \ddot{\mathbf{r}} \quad \mu = \frac{m M}{m + M}$$

$$\text{Si } M \gg m \quad \mu \simeq m$$

$\mathbf{F}_{sp}$ y $\mathbf{F}_{ps}$ son centrales $\Rightarrow L_{cm} = \text{cte}$

$$L_{cm} = \mathbf{r}_p \times \mathbf{p}_p + \mathbf{r}_s \times \mathbf{p}_s$$

$$L_{cm} = \left(\frac{M}{m+M} \right) \mathbf{r} \times m \left(\frac{M}{m+M} \right) \dot{\mathbf{r}} + \left(-\frac{M}{m+M} \right) \mathbf{r} \times M \left(-\frac{m}{m+M} \right) \dot{\mathbf{r}}$$

$$L_{cm} = \frac{m M}{m + M} \mathbf{r} \times \dot{\mathbf{r}}$$

$$L_{cm} = \mu \mathbf{r} \times \dot{\mathbf{r}}$$

$$\mathbf{r} = r \hat{\mathbf{r}}$$

$$\dot{\mathbf{r}} = \dot{r} \hat{\mathbf{r}} + r \dot{\theta} \hat{\boldsymbol{\theta}}$$

$$L_{cm} = \mu r \hat{r} \times (\dot{r} \hat{r} + r \dot{\theta} \hat{\theta})$$

$$L_{cm} = \mu r^2 \dot{\theta} \hat{n} \quad l = \mu r^2 \dot{\theta}$$

F_{sp} y F_{ps} son no conservativas

$$dW = F_{sp} dr_p + F_{ps} dr_s$$

$$= F_{sp} (dr_p - dr_s)$$

$$= F_{sp} d(\underbrace{r_p - r_s}_r)$$

$$= -\frac{G M m}{r^2} \hat{r} dr \hat{r}$$

$$dE_p = \frac{G m M}{r^2} dr \Rightarrow E_p = -\frac{G m M}{r} \quad \text{si: } E_p(r \rightarrow \infty) = 0$$

$$E_M = \frac{1}{2} m v_p^2 + \frac{1}{2} M v_s^2 - \frac{G M m}{r} = \text{cte}$$

$$E_M = \frac{1}{2} m \left(\frac{M}{m+M} \right)^2 \dot{r}^2 + \frac{1}{2} M \left(-\frac{m}{m+M} \right)^2 \dot{r}^2 - \frac{G M m}{r}$$

$$E_M = \frac{1}{2} \mu |\dot{r}|^2 - \frac{G M m}{r} = \frac{1}{2} \mu (\dot{r}^2 + r^2 \dot{\theta}^2) - \frac{G M m}{r}$$

$$E_M = \frac{1}{2} \mu \dot{r}^2 + \frac{1}{2} \mu r^2 \dot{\theta}^2 - \frac{G M m}{r}$$

$$E_M = \frac{1}{2} \mu \dot{r}^2 + \underbrace{\frac{1}{2} \frac{L^2}{\mu r^2}}_{\text{Potencial efectivo}} - \frac{G m M}{r}$$

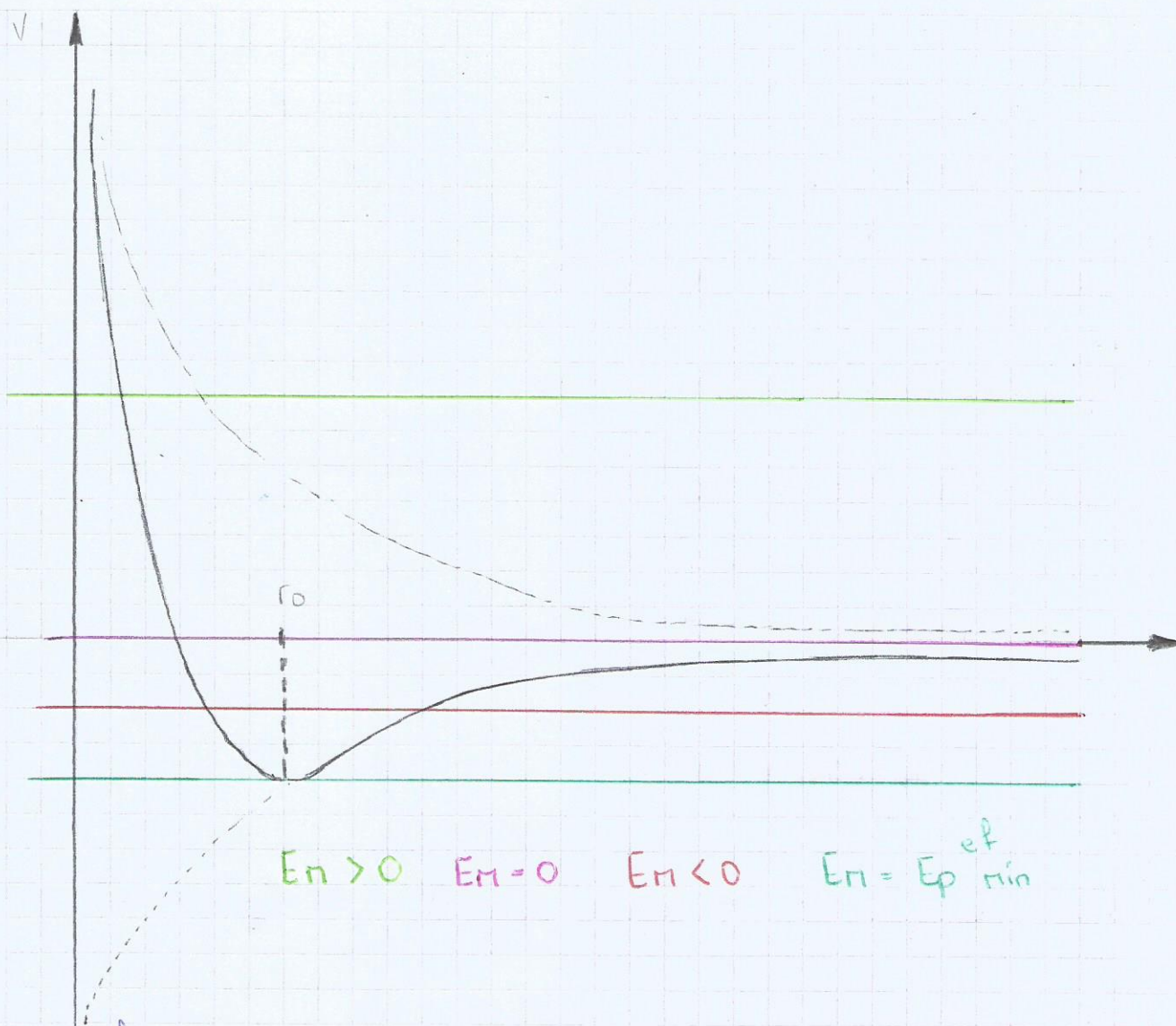
Potencial efectivo

$$E_{p\text{ef}} = E_{p\text{ct}} + E_{pg}$$

$$f_{\text{ct}} = -\frac{dE_{p\text{ct}}}{dr} \hat{r}$$

$$f_{\text{ct}} = \frac{L^2}{\mu r^3} \hat{r}$$

$$f_{\text{ct}} = \mu r \dot{\theta}^2 \hat{r}$$



$$E_{\min} > 0 \quad E_{\min} = 0 \quad E_{\min} < 0 \quad E_{\min} = E_p^{ef}$$

$$\frac{dE_p^{ef}}{dr} = -\frac{l^2}{Mr^3} + \frac{GMm}{r^2} = 0$$

$$r_0 = \frac{l^2}{GMm\mu} \quad E_p^{ef} = -\frac{1}{2} \frac{GM^2 m^2 \mu}{l^2}$$

$$\frac{d\theta}{dt} = \frac{l}{\mu r^2} = \frac{d\theta}{dr} \frac{dr}{dt} = \frac{d\theta}{dr} \dot{r}$$

$$r = \frac{k}{1 + e \cos \theta}$$

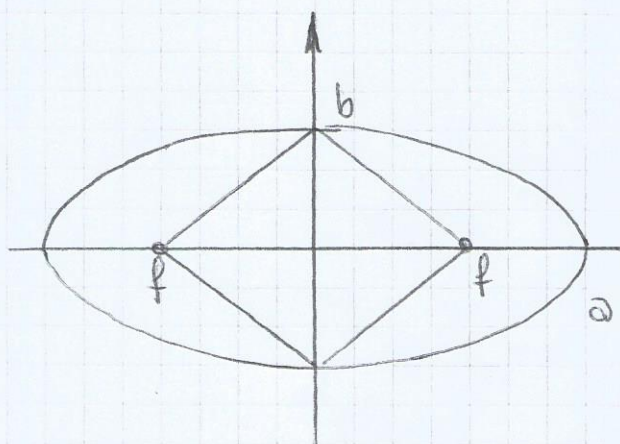
$$k = \frac{l^2}{Gm\mu M}$$

e excentricidad

$$e = \sqrt{1 + \frac{2kGM}{GmM}}$$

E	Órbita	Distancia focal (semieje)	Energía Mecánica
0	Circular	$a = b = k$	$E_{\text{el min}}$
$0 < E < 1$	Elíptica	$a = \frac{k}{1 - e^2}, b = \frac{k}{(1 - e)^2}$	$E_m < 0$
$E = 1$	Parabólica	$\frac{k}{2}$	$E_m = 0$
$E > 0$	Hiperbólica	$\frac{k}{1 + e}$	$E_m > 0$

Órbita elíptica:

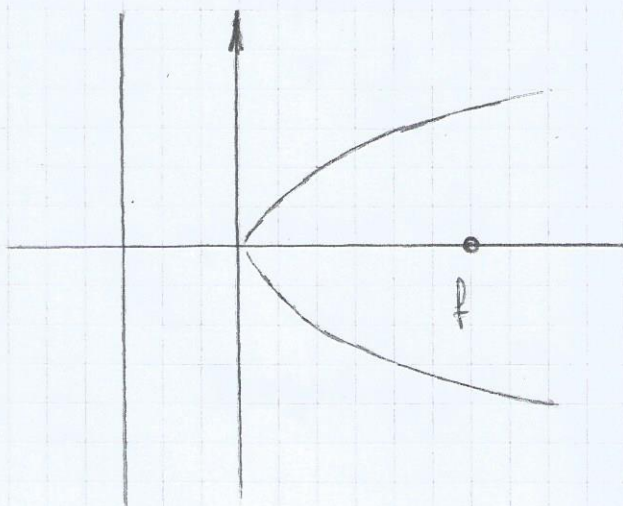


$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$e = \frac{f}{a}$$

$$b = \sqrt{a^2 - f^2}$$

Órbita parabólica:

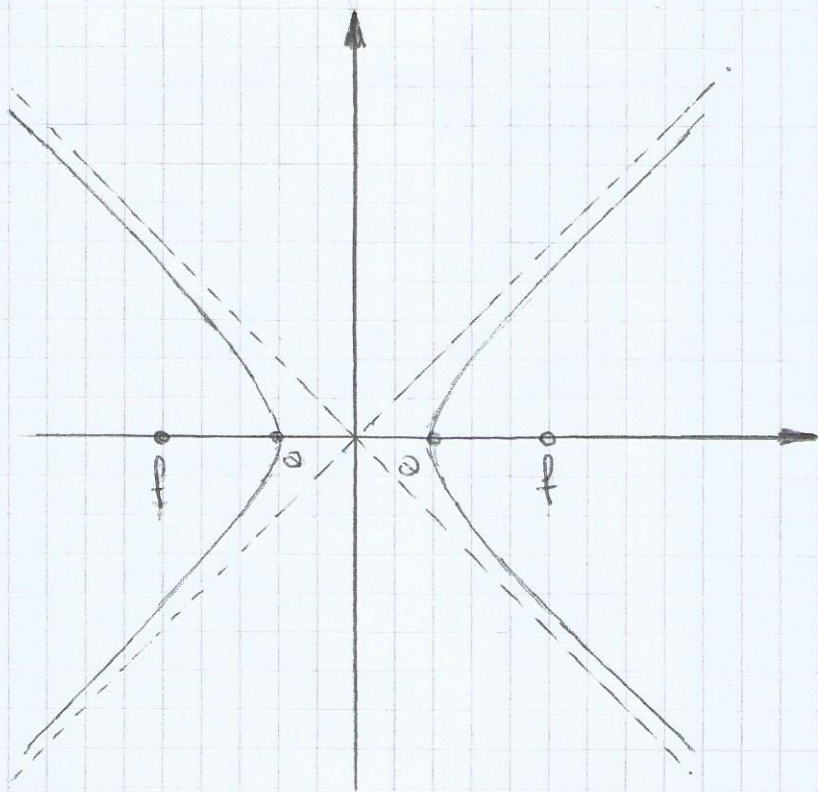


$$y^2 = 2fx$$

$$E = 1$$

directriz

Órbita hiperbólica:



$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$e = \frac{f}{a}$$

Período de las órbitas

$$A = \pi a b \quad (\text{elipse})$$

$$A = \pi a^2 (1 - e^2)^{1/2}$$

$$A = \pi a^2 \left(\frac{2k |E|}{G m M} \right)^{1/2}$$

$$a = \frac{k}{1 - e^2}$$

$$b = \frac{k}{(1 - e^2)^{1/2}}$$

Por otro lado:

(2^{da} ley de Kepler)

$$L_{\text{cm}} = \text{cte} \Rightarrow \frac{dA}{dt} = \frac{l}{2\mu} \Rightarrow A = \frac{l}{\mu} T$$

$$T = \frac{2\mu}{l} \cdot \pi a^2 \left(\frac{2k |E|}{G m M} \right)^{1/2} \Rightarrow l^2 = G m M k \mu$$

$$T^2 = \frac{4\pi^2 \mu a^3}{G m M} \quad (3^{\text{ra}} \text{ ley de Kepler})$$